

Asynchronous Transfer Mode (ATM) Fundamentals

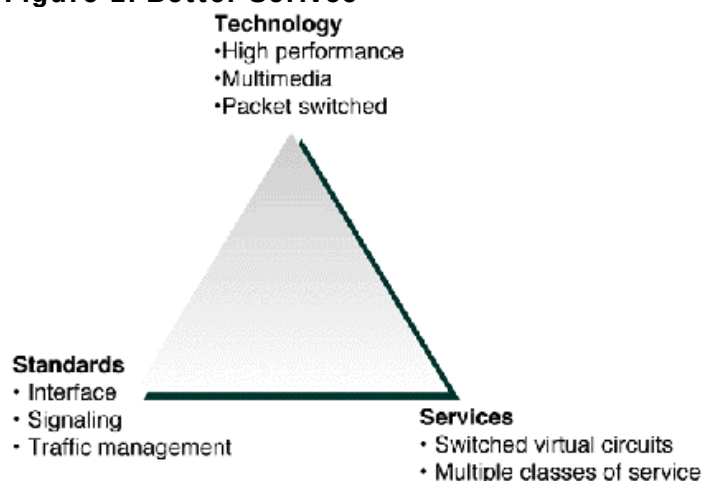
Definition

Asynchronous transfer mode (ATM) is a high-performance, cell-oriented switching and multiplexing technology that utilizes fixed-length packets to carry different types of traffic. ATM is a technology that will enable carriers to capitalize on a number of revenue opportunities through multiple ATM classes of services; high-speed local-area network (LAN) interconnection; voice, video, and future multimedia applications in business markets in the short term; and in community and residential markets in the longer term.

Overview

Changes in the structure of the telecommunications industry and market conditions have brought new opportunities and challenges for network operators and public service providers. Networks that have been primarily focused on providing better voice services are evolving to meet new multimedia communications challenges and competitive pressures. Services based on asynchronous transfer mode (ATM) and synchronous digital hierarchy (SDH)/synchronous optical network (SONET) architectures provide the flexible infrastructure essential for success in this evolving market (see *Figure 1*).

Figure 1. Better Service



ATM, which was once envisioned as the technology of future public networks, is now a reality, with service providers around the world introducing and rolling out ATM and ATM-based services. The ability to exploit the benefits of ATM technology within the public network successfully will provide strategic competitive advantage to carriers and enterprises alike.

In addition to revenue opportunities, ATM reduces infrastructure costs through efficient bandwidth management, operational simplicity, and the consolidation of overlay networks. Carriers can no longer afford to go through the financial burden and time required to deploy a separate network for each new service requirement (e.g., dedicating a network for a single service such as transparent LAN or frame relay). ATM technology will allow core network stability while allowing service interfaces and other equipment to evolve rapidly.

Topics

1. Definition of ATM
2. Benefits of ATM
3. ATM Technology
4. ATM Services
5. ATM Standards
6. ATM LAN Emulation
7. Voice over ATM
8. Video over ATM
9. ATM Traffic Management
10. ATM Applications
11. Nortel's ATM Vision

Self-Test

Correct Answers

Glossary

1. Definition of ATM

Asynchronous transfer mode (ATM) is a technology that has its history in the development of broadband ISDN in the 1970s and 1980s. Technically, it can be viewed as an evolution of packet switching. Like packet switching for data (e.g., X.25, frame relay, transmission control protocol [TCP]/Internet protocol [IP], ATM integrates the multiplexing and switching functions, is well suited for bursty traffic (in contrast to circuit switching), and allows communications between

devices that operate at different speeds. Unlike packet switching, ATM is designed for high-performance multimedia networking. ATM technology has been implemented in a very broad range of networking devices:

- PC, workstation, and server network interface cards
- switched-Ethernet and token-ring workgroup hubs
- workgroup and campus ATM switches
- ATM enterprise network switches
- ATM multiplexers
- ATM–edge switches
- ATM–backbone switches

ATM is also a capability that can be offered as an end-user service by service providers (as a basis for tariffed services) or as a networking infrastructure for these and other services. The most basic service building block is the ATM virtual circuit, which is an end-to-end connection that has defined end points and routes but does not have bandwidth dedicated to it. Bandwidth is allocated on demand by the network as users have traffic to transmit. ATM also defines various classes of service to meet a broad range of application needs.

ATM is also a set of international interface and signaling standards defined by the International Telecommunications Union–Telecommunications (ITU–T) Standards Sector (formerly the CCITT). The ATM Forum has played a pivotal role in the ATM market since its formulation in 1991. The ATM Forum is an international voluntary organization composed of vendors, service providers, research organizations, and users. Its purpose is to accelerate the use of ATM products and services through the rapid convergence of interoperability specifications, promotion of industry cooperation, and other activities. Developing multivendor implementation agreements also furthers this goal.

2. Benefits of ATM

The benefits of ATM are the following:

- high performance via hardware switching
- dynamic bandwidth for bursty traffic
- class-of-service support for multimedia

- scalability in speed and network size
- common LAN/WAN architecture
- opportunities for simplification via VC architecture
- international standards compliance

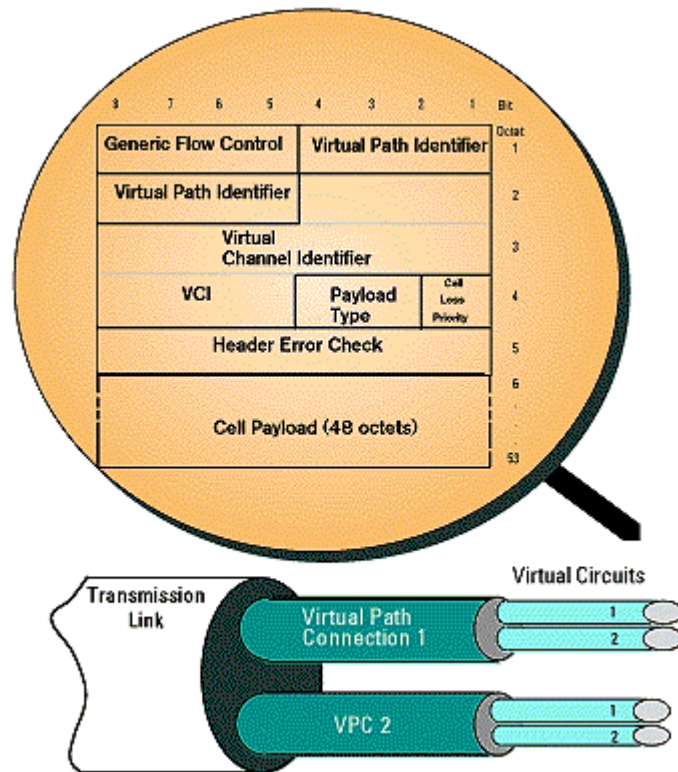
The high-level benefits delivered through ATM services deployed on ATM technology using international ATM standards can be summarized as follows:

- high performance via hardware switching with terabit switches on the horizon
- dynamic bandwidth for bursty traffic meeting application needs and delivering high utilization of networking resources; most applications are or can be viewed as inherently bursty; data applications are LAN-based and are very bursty, voice is bursty, as both parties are neither speaking at once nor all the time; video is bursty, as the amount of motion and required resolution varies over time
- class-of-service support for multimedia traffic allowing applications with varying throughput and latency requirements to be met on a single network
- scalability in speed and network size supporting link speeds of T1/E1 to OC-12 (622 Mbps) today and into the multi-Gbps range before the end of the decade; networks that scale to the size of the telephone network (i.e., as required for residential applications) are envisaged
- common LAN/WAN architecture allowing ATM to be used consistently from one desktop to another; traditionally, LAN and WAN technologies have been very different, with implications for performance and interoperability
- opportunities for simplification via switched VC architecture; this is particularly for LAN-based traffic that today is connectionless in nature; the simplification possible through ATM VCs could be in areas such as billing, traffic management, security, and configuration management
- international standards compliance in central-office and customer-premises environments allowing for multivendor operation

3. ATM Technology

In ATM networks, all information is formatted into fixed-length cells consisting of 48 bytes (8 bits per byte) of payload and 5 bytes of cell header (see *Figure 2*). The fixed cell size ensures that time-critical information such as voice or video is not adversely affected by long data frames or packets. The header is organized for efficient switching in high-speed hardware implementations and carries payload-type information, virtual-circuit identifiers, and header error check.

Figure 2. Fixed-Length Cells



ATM is connection oriented. Organizing different streams of traffic in separate calls allows the user to specify the resources required and allows the network to allocate resources based on these needs. Multiplexing multiple streams of traffic on each physical facility (between the end user and the network or between network switches)—combined with the ability to send the streams to many different destinations—enables cost savings through a reduction in the number of interfaces and facilities required to construct a network.

ATM standards defined two types of ATM connections: virtual path connections (VPCs), which contain virtual channel connections (VCCs). A virtual channel connection (or virtual circuit) is the basic unit, which carries a single stream of cells, in order, from user to user. A collection of virtual circuits can be bundled

together into a virtual path connection. A virtual path connection can be created from end-to-end across an ATM network. In this case, the ATM network does not route cells belonging to a particular virtual circuit. All cells belonging to a particular virtual path are routed the same way through the ATM network, thus resulting in faster recovery in case of major failures.

An ATM network also uses virtual paths internally for the purpose of bundling virtual circuits together between switches. Two ATM switches may have many different virtual channel connections between them, belonging to different users. These can be bundled by the two ATM switches into a virtual path connection. This can serve the purpose of a virtual trunk between the two switches. This virtual trunk can then be handled as a single entity by, perhaps, multiple intermediate virtual path cross connects between the two virtual circuit switches.

Virtual circuits can be statically configured as permanent virtual circuits (PVCs) or dynamically controlled via signaling as switched virtual circuits (SVCs). They can also be point-to-point or point-to-multipoint, thus providing a rich set of service capabilities. SVCs are the preferred mode of operation because they can be dynamically established, thus minimizing reconfiguration complexity.

4. ATM Classes of Services

ATM is connection oriented and allows the user to specify the resources required on a per-connection basis (per SVC) dynamically. There are the five classes of service defined for ATM (as per ATM Forum UNI 4.0 specification). The QoS parameters for these service classes are summarized in *Table 1*.

Table 1. ATM Service Classes

Service Class	Quality of Service Parameter
constant bit rate (CBR)	This class is used for emulating circuit switching. The cell rate is constant with time. CBR applications are quite sensitive to cell-delay variation. Examples of applications that can use CBR are telephone traffic (i.e., nx64 kbps), videoconferencing, and television.
variable bit rate–non-real time (VBR–NRT)	This class allows users to send traffic at a rate that varies with time depending on the availability of user information. Statistical multiplexing is provided to make optimum use of network resources. Multimedia e-mail is an example of VBR–NRT.
variable bit rate–real time (VBR–RT)	This class is similar to VBR–NRT but is designed for applications that are sensitive to cell-delay variation. Examples for real-time VBR are voice with speech activity detection (SAD) and interactive compressed video.
available bit rate (ABR)	This class of ATM services provides rate-based flow control and is aimed at data traffic such as file transfer and e-mail. Although the standard does not require the cell transfer delay and cell-loss ratio to be guaranteed or minimized, it is desirable for switches to minimize delay and loss as much as possible. Depending upon the state of

	congestion in the network, the source is required to control its rate. The users are allowed to declare a minimum cell rate, which is guaranteed to the connection by the network.
unspecified bit rate (UBR)	This class is the catch-all, other class and is widely used today for TCP/IP.

The ATM Forum has identified the following technical parameters to be associated with a connection. These terms are outlined in *Table 2*.

Table 2. ATM Technical Parameters

Technical Parameter	Definition
cell loss ratio (CLR)	CLR is the percentage of cells not delivered at their destination because they were lost in the network due to congestion and buffer overflow.
cell transfer delay (CTD)	The delay experienced by a cell between network entry and exit points is called CTD. It includes propagation delays, queuing delays at various intermediate switches, and service times at queuing points.
cell delay variation (CDV)	CDV is a measure of the variance of the cell transfer delay. High variation implies larger buffering for delay- sensitive traffic such as voice and video.
peak cell rate (PCR)	The maximum cell rate at which the user will transmit PCR is the inverse of the minimum cell interarrival time.
sustained cell rate (SCR)	This is the average rate, as measured over a long interval, in the order of the connection lifetime.
burst tolerance (BT)	This parameter determines the maximum burst that can be sent at the peak rate. This is the bucket-size parameter for the enforcement algorithm that is used to control the traffic entering the network.

Finally, there are a number of ATM classes of service. These classes are all outlined in *Table 3*.

Table 3. ATM Classes of Services

Class of Service	CBR	VBR-NRT	VBR-RT	ABR	UBR
CLR	yes	yes	yes	yes	no
CTD	yes	no	yes	no	no
CDV	yes	yes	yes	no	no
PCR	yes	yes	yes	no	yes
SCR	no	yes	yes	no	no
BT @ PCR	no	yes	yes	no	no
flow control	no	no	no	yes	no

Its extensive class-of-service capabilities make ATM the technology of choice for multimedia communications.

5. ATM Standards

The ATM Forum has identified a cohesive set of specifications that provide a stable ATM framework. The first and most basic ATM standards are those that provide the end-to-end service definitions as described in *Topic 4*. An important ATM standard and service concept is that of service interworking between ATM and frame relay (a fast-growing pervasive service), whereby ATM services can be seamlessly extended to lower-speed frame-relay users. Frame relay is a network technology that is also based on virtual circuits using variable-length frame transmission between users.

ATM user network interface (ATM UNI) standards specify how a user connects to the ATM network to access these services. A number of standards have been defined for T1/E1, 25 Mbps, T3/E3, OC-3 (155 Mbps) and OC-12 with OC-48 (2.4 Gbps) in the works. OC-3 interfaces have been specified for use over single-mode fiber (for wide-area applications) and over unshielded twisted pair or multimode fiber for lower-cost, in-building applications.

The following two ATM networking standards have been defined that provide connectivity between network switches and between networks:

- broadband intercarrier interface (B-ICI)
- public network-to-network interface (P-NNI)

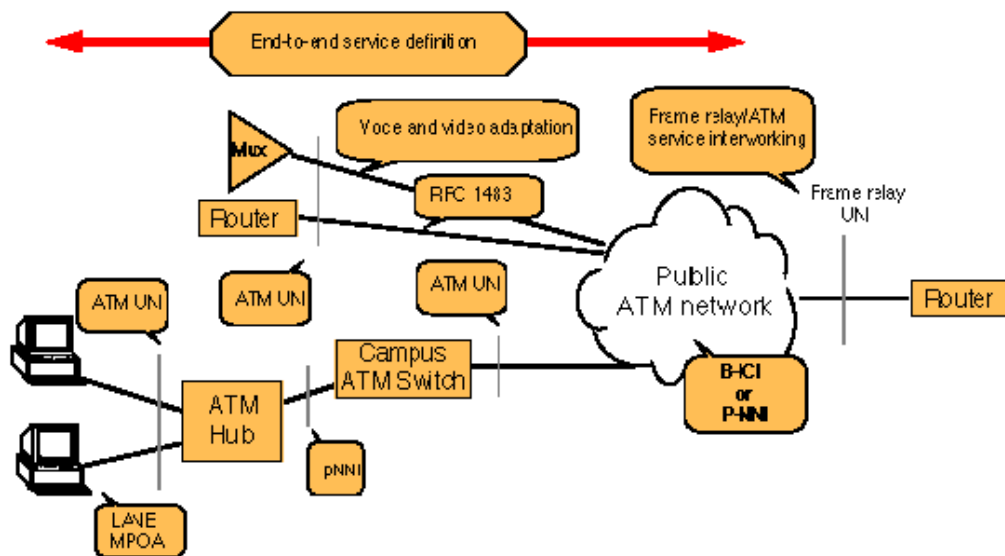
P-NNI is the more feature-rich of the two and supports class of service-sensitive routing and bandwidth reservation. It provides topology-distribution mechanisms based on advertisement of link metrics and attributes, including bandwidth metrics. It uses a multilevel hierarchical routing model providing scalability to large networks. Parameters used as part of the path-computation process include the destination ATM address, traffic class, traffic contract, QoS requirements and link constraints. Metrics that are part of the ATM routing system are specific to the traffic class and include quality of service-related metrics (e.g., CTD, CLR) and bandwidth-related metrics (e.g., PCR). The path computation process includes overall network-impact assessment, avoidance of loops, minimization of rerouting attempts, and use of policy (inclusion/exclusion in rerouting, diverse routing, and carrier selection). Connection admission controls (CACs) define procedures used at the edge of the network, whereby the call is accepted or rejected based the ability of the network to support the requested QoS. Once a VC has been established across the network, network resources have to be held and quality service guaranteed for the duration of the connection.

All ATM traffic is carried in cells, yet no applications use cells. So, specific ways of putting the data into cells are defined to enable the receiver to reconstruct the

original traffic. Three important schemes are highlighted in *Figure 3* and discussed in detail later in the tutorial.

- RFC1483, which specifies how interrouter traffic is encapsulated into ATM using ATM adaptation Layer 5 (AAL-5); AAL-5 is optimized for handling framed traffic and has similar functionality to that provided by HDLC framing in frame relay, SDLC, and X.25
- ATM LAN emulation (LANE) and multiprotocol over ATM (MPOA), which are designed to support dynamic use of ATM SVCs primarily for TCP/IP; LANE, which is a current standard that is widely deployed and will be a subset of the MPOA standard (which is targeted for standardization only in mid-1997), will be discussed later in the tutorial
- voice and video adaptation schemes that can use AAL-1, which is defined for high efficiency—for traffic that itself has no natural breaks, such as a circuit carrying bits at a fixed rate

Figure 3. Data Insertion in Cells

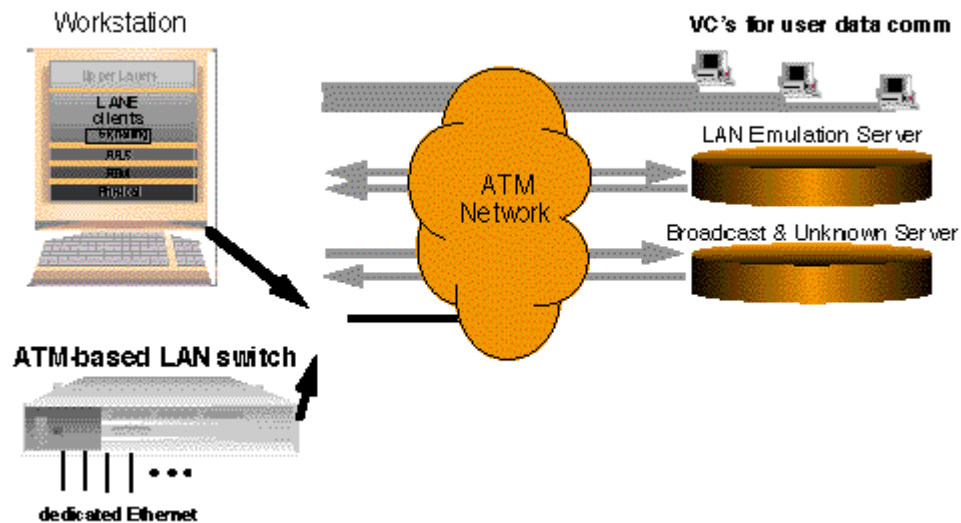


6. ATM LAN Emulation

ATM-based Ethernet switches and ATM workgroup switches are being deployed by end users at various corporate sites. The most widely used set of standards in local ATM environments is ATM LAN emulation (LANE) (see *Figure 4*). ATM LAN emulation is used to make the ATM SVC network appear to be a collection of virtual-Ethernet/IEEE 802.3 and token-ring/IEEE802.5 LANs. The replication of most of the characteristics of existing LANs means that LAN emulation enables

existing LAN applications to run over ATM transparently, this latter characteristic leading to its wide deployment. In ATM LAN emulation, most unicast LAN traffic moves directly between clients over direct ATM SVCs, while multicast traffic is handled via a server functionality. Bridging is used to interconnect real LANs and emulated LANs running on ATM, while routing is used to interconnect ATM-emulated LANs and other WAN or LAN media for purposes of routing scalability, protocol spoofing, or security firewalls.

Figure 4. ATM LAN Emulation (LANE)



The ATM Forum LANE implementation agreement specifies two types of LANE network components connected to an ATM network.

- LANE clients which function as end systems, such as computers with ATM interfaces that operate as file servers; end-user workstations or personal computers; Ethernet or token-ring switches that support ATM networking; and routers, bridges, and ATM ENS with membership in an emulated ATM LAN
- LANE servers that support ATM LANE service for configuration management, multicast support, and address resolution

The LAN-emulation service may be implemented in the same devices as clients or involve other ATM network devices. The communications interface, LAN emulation user-network interface (LUNI), is the sequence and contents of the messages that the clients ultimately use to transfer traffic of the type expected on IEEE 802.3/5 LANs. The component of the LAN-emulation service that deals with initialization (i.e., emulates plugging the terminal into a LAN hub), is the LAN emulation configuration server (LECS). It directs a client to connect to a

particular LAN emulation server (LES). The LES is the component of the LAN-emulation service that performs the address registration and resolution. The LES is responsible for mapping IEEE 48-bit MAC addresses and token-ring route descriptors to ATM addresses. One very important MAC address for clients is the MAC-layer broadcast address that is used to send traffic to all locations on a LAN. In LAN emulation, this function is performed by the broadcast and unknown server (BUS). ATM LANE is a comprehensive set of capabilities which has been widely deployed in ATM networks.

ATM LANE is an element of the multiple protocol over ATM (MPOA) architecture that is being defined by the ATM Forum. This work is addressing encapsulation of multiple protocols over ATM, automatic address resolution, and the routing issues associated with minimizing multiple router hops in ATM networks.

7. Voice over ATM

As real-time voice services have been traditionally supported in the WAN via circuit-based techniques (e.g., via T1 multiplexers or circuit switching), it is natural to map these circuits to ATM CBR PVCs using circuit emulation and ATM adaptation Layer 1 (AAL1). However, there are significant disadvantages in using circuit emulation in that the bandwidth must be dedicated for this type of traffic (whether there is useful information being transmitted or not), providing a disincentive for corporate users to implement circuit emulation as a long-term strategy. For example, a T1 1.544-Mbps circuit requires 1.74 Mbps of ATM bandwidth when transmitted in circuit-emulation mode. This does not downplay its importance as a transitional strategy to address the installed base.

As technology has evolved, the inherent burstiness of voice and many real-time applications can be exploited (along with sophisticated compression schemes) to decrease the cost of transmission significantly through the use of VBR-RT connections over ATM.

VBR techniques for voice exploit the inherently bursty nature of voice communication, as there are silence periods that can result in increased efficiency. These following silence periods (in decreasing levels of importance) arise:

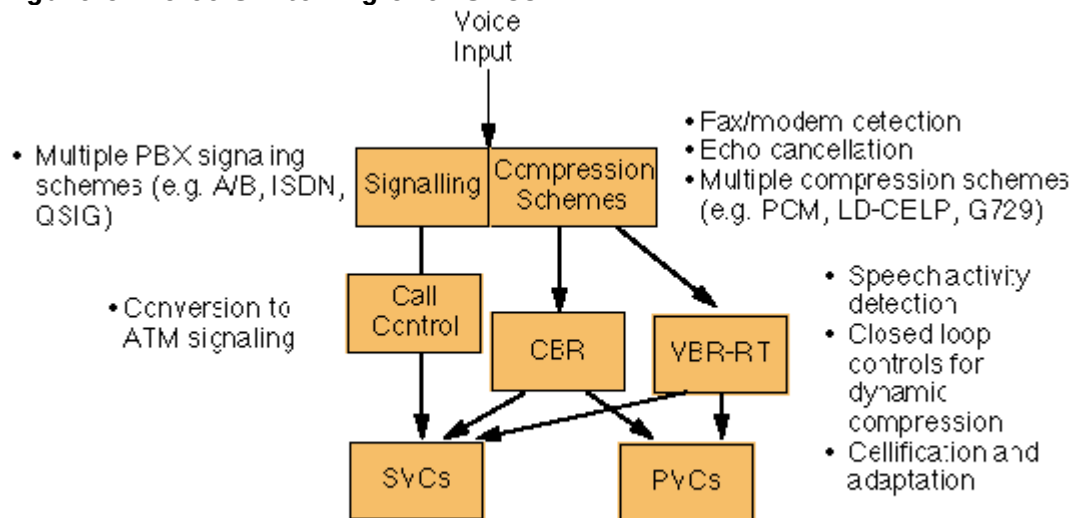
- when no call is up on a particular trunk; that is, the trunk is idle during off-peak hours (trunks are typically engineered for a certain call-blocking probability: at night, all the trunks could be idle)
- when the call is up, but only one person is talking at a given time
- when the call is up, and no one is talking

Work is just starting in the ATM Forum on ATM adaptation for VBR voice.

The addition of more bandwidth-effective voice coding (e.g., standard voice is coded using 64-kbps PCM) is economically attractive, particularly over long-haul circuits and T1 ATM interfaces. Various compression schemes have been standardized in the industry (e.g., G720 series of standards). Making these coding schemes dynamic provides the network operator the opportunity to free up bandwidth under network-congestion conditions. For example, with the onset of congestion, increased levels of voice compression could be dynamically invoked, thus freeing up bandwidth and potentially alleviating the congestion while diminishing the quality of the voice during these periods.

A further enhancement to the support of voice over ATM is to support voice switching over SVCs. This entails interpreting PBX signaling and routing voice calls to the appropriate destination PBX (see *Figure 5*). The advantage from a traffic management perspective is that connection admission controls can be applied to new voice calls; under network congestion conditions, these calls could be rerouted over the public network and therefore not cause additional levels of congestion.

Figure 5. Voice Switching over SVCs



The ATM Forum is currently focusing its efforts on voice handled on CBR SVCs. VBR-RT voice is a future standards activity.

8. Video over ATM

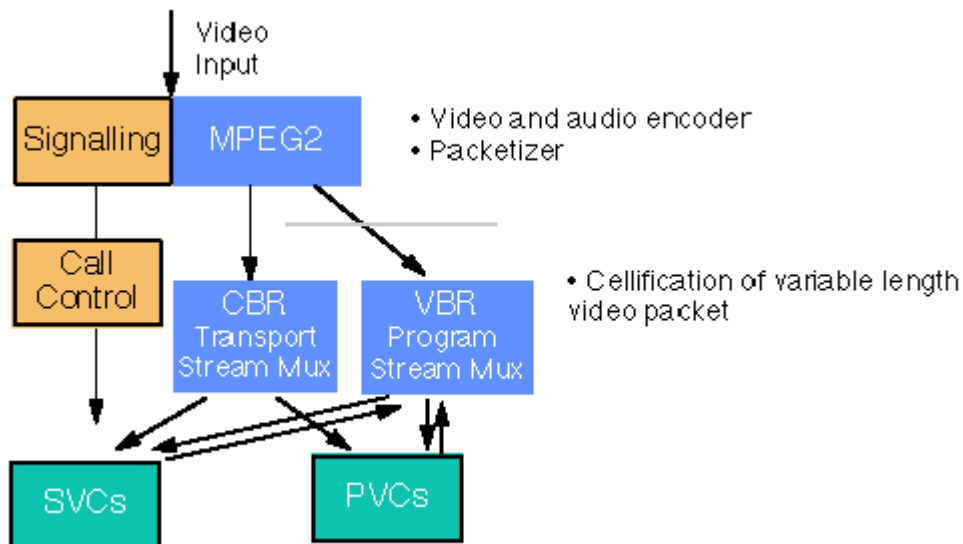
While circuit-based videoconferencing streams (including motion JPEG running at rates around 10 Mbps) can be handled by standard circuit emulation using

AAL-1, the ATM Forum has specified the use of VBR-RT VCs using AAL-5 for MPEG2 on ATM for video-on-demand applications, as this approach makes better use of networking resources.

MPEG is a set of standards addressing coding of video and surround-sound audio signals and synchronization of video and audio signals during the playback of MPEG data. It runs in the 2 Mbps to 15 Mbps range (with bursts above these rates) corresponding to VCR and broadcast quality respectively. The initial MPEG standard (MPEG1) was targeted at VHS-quality video and audio. MPEG2 targets applications requiring broadcast-quality video and audio and HDTV. MPEG2 coding can result in one of the following two modes:

- **program streams**—variable-length packets that carry a single program or multiple programs with a common time base
- **transport streams**—188-byte packets that contain multiple programs (for examples, see *Figure 6*).

Figure 6. Transport Streams



In both cases, time stamps are inserted into MPEG2 packets during the encoding and multiplexing process. MPEG2 assumes a constant-delay model across the network, thus allowing the decoder to exactly follow the original encoder source clock. Due to the cost of coding, MPEG2 is primarily used in a non-interactive broadcast mode as would be the case for a point-to-multipoint broadcast in residential video on demand applications and in a business TV application for training or employee communications.

9. ATM Traffic Management

Broadly speaking, the objectives of ATM traffic management are to deliver quality-of-service (QoS) guarantees for the multimedia applications and provide overall optimization of network resources. Meeting these objectives enables enhanced classes of service and offers the potential for service differentiation and increased revenues, while simplifying network operations and reducing network cost.

ATM traffic management and its various functions can be categorized into three distinct elements based on timing requirements.

First, are nodal-level controls that operate in real time. These are implemented in hardware and include queues supporting different loss and delay priorities, fairly weighted queue-servicing algorithms, and rate controls that provide policing and traffic shaping. Well-designed switch-buffer architectures and capacity are critical to effective network operation. Actual network experience and simulation has indicated that large, dynamically allocated output buffers provide the flexibility to offer the best price performance for supporting various traffic types with guaranteed QoS. Dynamically managing buffer space means that all shared buffer space is flexibly allocated to VCs on an as-needed basis. Additionally, per virtual connection (VC) queuing enables traffic shaping, and early and partial packet-level discard have been shown to improve network performance significantly.

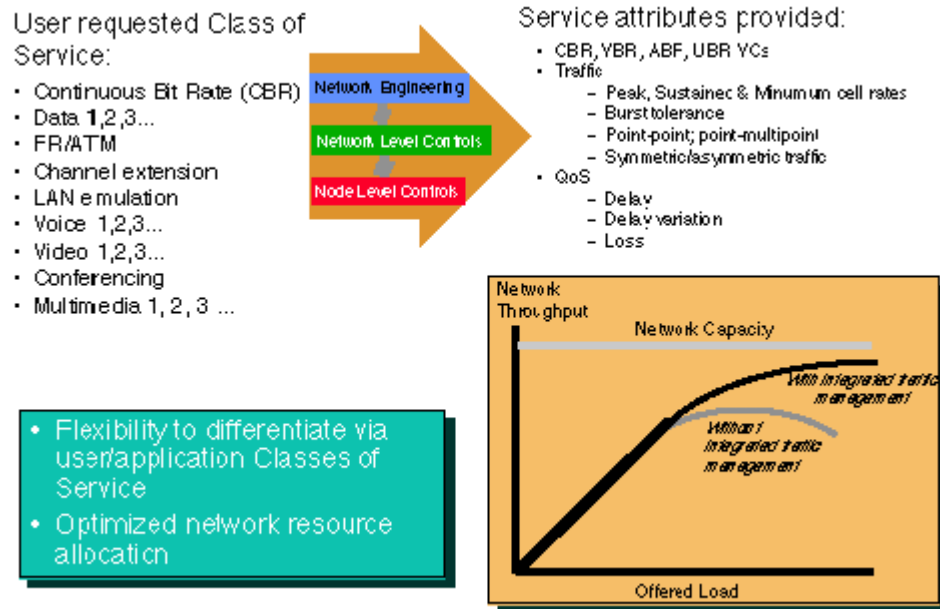
Second, network-level controls operate in near real time. These are typically, but not exclusively, implemented in software including connection admission control (CAC) for new connections, network routing and rerouting systems, and flow-control-rate adaptation schemes. Network-level controls are the heart of any traffic-management system. Connection admission controls support sophisticated equivalent-bandwidth algorithms with a high degree of configuration flexibility, based on the cell rate for CBR VCs, average cell rate plus a configurable increment for VBR VCs, and minimum cell rate for ABR VCs. Dynamic class-of-service routing standards define support for fully distributed link-state routing protocols, auto-reconfiguration on failure and on congestion, and dynamic load spreading on trunk groups.

Flow control involves adjusting the cell rate of the source in response to congestion conditions and requires the implementation of closed loop congestion mechanisms. This does not apply to CBR traffic. For VBR and UBR traffic, flow control is left as a CPE function. With ABR, resource management (RM) cells are defined, which allow signaling of the explicit rate to be used by traffic sources. This is termed rate-based flow control. ABR is targeted at those applications that do not have fixed or predictable bandwidth requirements and require access to any spare bandwidth as quickly as possible while experiencing very low cell loss. This allows network operators to maximize the bandwidth utilization of their

network and sell spare capacity to users at a substantial discount while still providing QoS guarantees. To enhance the effectiveness of network-resource utilization, the ABR standard provides for end-to-end, segment-by-segment, and hop-by-hop service adaptation.

Third, network engineering capabilities operating in nonreal time support data collection, configuration management, and planning tools (see *Figure 7*).

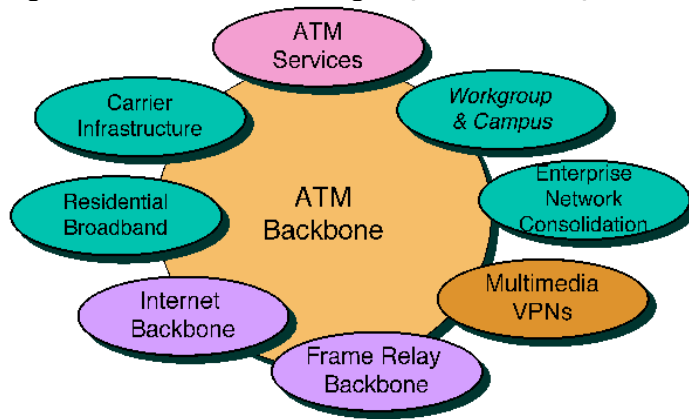
Figure 7. Network Engineering Capabilities



10. ATM Applications

ATM technologies, standards, and services are being applied in a wide range of networking environments, as described briefly below (see *Figure 8*).

Figure 8. ATM Technologies, Standards, and Services



- **ATM services**—Service providers globally are introducing or already offering ATM services to their business users.
- **ATM workgroup and campus networks**—Enterprise users are deploying ATM campus networks based on the ATM LANE standards. Workgroup ATM is more of a niche market with the wide acceptance of switched-Ethernet desktop technologies.
- **ATM enterprise network consolidation**—A new class of product has evolved as an ATM multimedia network-consolidation vehicle. It is called an ATM enterprise network switch. A full-featured ATM ENS offers a broad range of in-building (e.g., voice, video, LAN, and ATM) and wide-area interfaces (e.g., leased line, circuit switched, frame relay, and ATM at narrowband and broadband speeds) and supports ATM switching, voice networking, frame-relay SVCs, and integrated multiprotocol routing.
- **multimedia virtual private networks and managed services**—Service providers are building on their ATM networks to offer a broad range of services. Examples include managed ATM, LAN, voice and video services (these being provided on a per-application basis, typically including customer-located equipment and offered on an end-to-end basis), and full-service virtual private-networking capabilities (these including integrated multimedia access and network management).
- **frame-relay backbones**—Frame-relay service providers are deploying ATM backbones to meet the rapid growth of their frame-relay services to use as a networking infrastructure for a range of data services and to enable frame relay to ATM service interworking services.

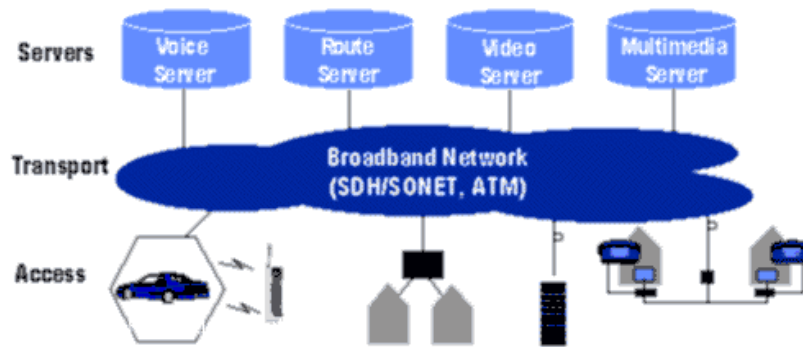
- **Internet backbones**—Internet service providers are likewise deploying ATM backbones to meet the rapid growth of their frame-relay services, to use as a networking infrastructure for a range of data services, and to enable Internet class-of-service offerings and virtual private intranet services.
- **residential broadband networks**—ATM is the networking infrastructure of choice for carriers establishing residential broadband services, driven by the need for highly scalable solutions.
- **carrier infrastructures for the telephone and private-line networks**—Some carriers have identified opportunities to make more-effective use of their SONET/SDH fiber infrastructures by building an ATM infrastructure to carry their telephony and private-line traffic.

11. Nortel's ATM Vision

Nortel believes that ATM is the only viable backbone networking technology that can meet the objective of making multimedia calls as easy, reliable, and secure as voice calls are today.

ATM, coupled with SONET/SDH for fiber transport, sits at the core of Nortel's long-term architectural vision. That vision embraces various residential, business, and mobile access arrangements with a set of voice/data/video and, ultimately, multimedia servers. There will be many ways of accessing ATM networks including desktop ATM, switched Ethernet, wireless, and xDSL, to name a few. The vision includes extensive support of multiple classes of service for native ATM, IP-based, frame-relay-based, and circuit-based applications. ATM accommodates the inherently bursty nature of data, voice, and video applications and the compressibility of these traffic types for increased storage and bandwidth effectiveness. Nortel also believes that frame relay and ATM, being both virtual-circuit based, provide a service continuum supporting the broadest sets of speeds from sub-64 kbps all the way to Gbps. Finally, Nortel envisages a family of application servers around the periphery of this network to provide a range of data, image, video and voice services that take advantage of increasing insensitivity of the network to distance (see *Figure 9*).

Figure 9. Nortel's ATM Architectural Vision



A flexible scalable architecture:

- Service and media independent access
- Multiservice high performance transport
- Application servers on industry standard platforms
- Bandwidth on demand eliminating network as bottleneck

Self-Test

1. Which method is better suited to handle bursty traffic?
 - a. circuit switching
 - b. ATM
2. Which organization is currently responsible for international signaling and interface standards?
 - a. ITU
 - b. CCITT
1. In ATM networks all information is formatted into fixed-length cells consisting of _____ bytes.
 - a. 48
 - b. 64
4. The basic connection unit in an ATM network is known as the _____.
 - a. virtual channel connection
 - b. virtual path connection

5. Which virtual circuit connection is the preferred mode of operation in an ATM network?
 - a. permanent virtual circuits
 - b. switched virtual circuits
6. CBR is used for _____.
 - a. multimedia e-mail
 - b. videoconferencing
7. Which ATM networking standard supports the greater range of features?
 - a. P–NNI
 - b. B–ICI
8. In ATM LAN emulation, most unicast LAN traffic moves directly between clients over _____.
 - a. servers
 - b. direct ATM SVCs
9. The initial MPEG standard (MPEG1) was targeted at _____.
 - a. VHS–quality video and audio
 - b. broadcast-quality video and audio
10. ATM timing requirements feature _____ elements.
 - a. 3
 - b. 4
 - c. 64

Correct Answers

1. Which method is better suited to handle bursty traffic?
 - a. circuit switching
 - b. ATM**

See Topic 1.

2. Which organization is currently responsible for international signaling and interface standards?

a. ITU

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See Topic 1.

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4. The basic connection unit in an ATM network is known as the _____.

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b. switched virtual circuits

See Topic 3.

6. CBR is used for _____.

a. multimedia e-mail

b. videoconferencing

See Topic 6.

7. Which ATM networking standard supports the greater range of features?

a. P-NNI

b. B-ICI

See Topic 5.

8. In ATM LAN emulation, most unicast LAN traffic moves directly between clients over _____.

a. servers

b. direct ATM SVCs

See Topic 8.

9. The initial MPEG standard (MPEG1) was targeted at _____.

a. VHS-quality video and audio

b. broadcast-quality video and audio

See Topic 8.

10. ATM timing requirements feature _____ elements.

a. 3

b. 4

c. 64

See Topic 9.

Glossary

AAL-1

ATM adaption Layer 1

AAL-2

ATM adaption Layer 2

AAL-3

ATM adaption Layer 3

AAL-4

ATM adaption Layer 4

AAL-5

ATM adaption Layer 5

ABR

available bit rate

ATM

asynchronous transfer mode

ATM UNI

ATM user network interface

BT

burst tolerance

CAC

connection admission control

CBR

constant bit rate

CCITT

Comite Consultif Internationale de Telegraphique et Telephonique

CLR

cell loss ratio

CDV

cell delay variation

CTD

cell transfer delay

IEEE

Institute of Electrical and Electronic Engineers

ITU-T

International Telecommunications Union–Telecommunications Standards Sector

LAN

local-area network

LANE

ATM LAN emulation

LES

LAN emulation server

LUNI

LAN emulation user-network interface

MPOA

multiple protocol over ATM

PCR

peak cell rate

P–NNI

public network-to-network interface

PVC

permanent virtual circuit

RM

resource management

SAD

speech activity detection

SCR

sustained cell rate

SDH/SONET

synchronous digital hierarchy/synchronous optical network

SVC

switched virtual circuit

TCP/IP

transmission control protocol/Internet protocol

UBR

unspecified bit rate

VBR–NRT

variable bit rate–nonreal time

VCC

virtual-channel connections

VPC

virtual-path connections